

# Programming and Databases

Joern Ploennigs

Overview

## INTRODUCTION



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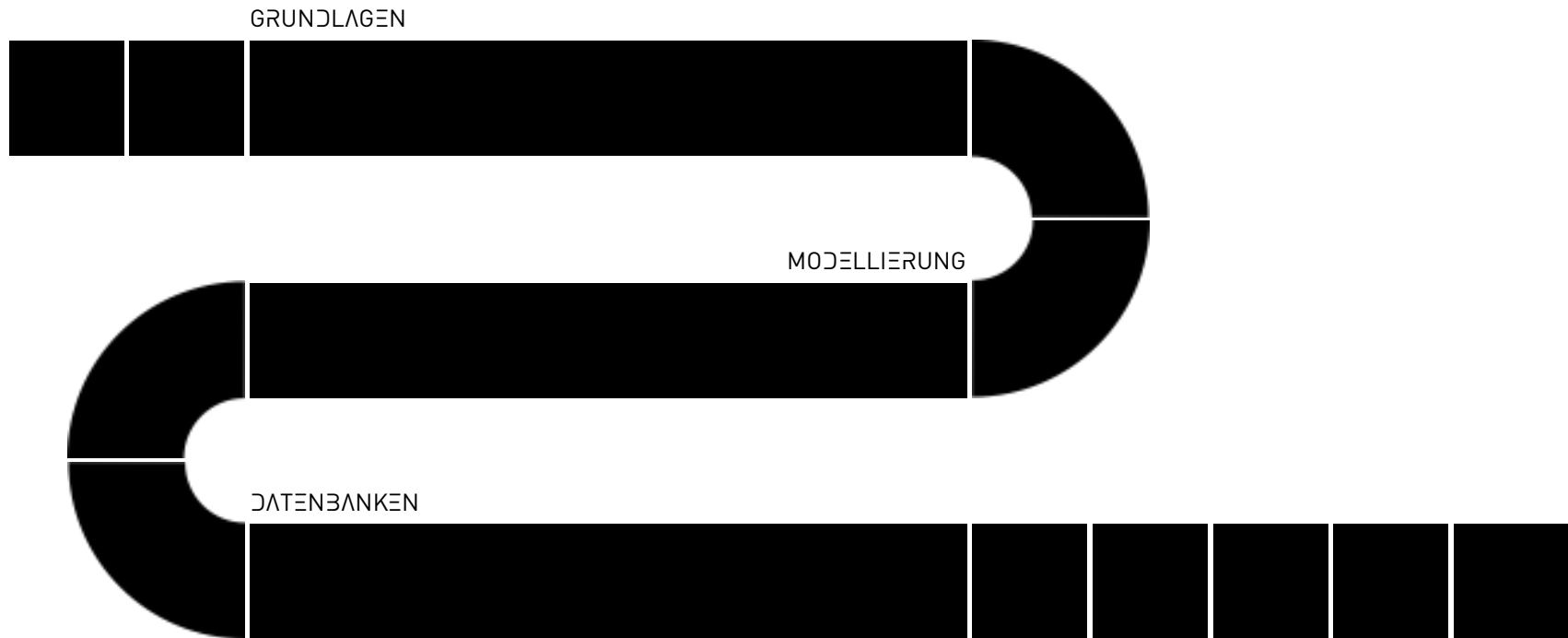
Phone: 0381 498-3500

## OBJECTIVES

- Develop an understanding of and proficiency in the fundamentals of programming and databases
- Acquire practical skills to solve engineering problems using software
- Familiarize yourself with current approaches and technologies in software development

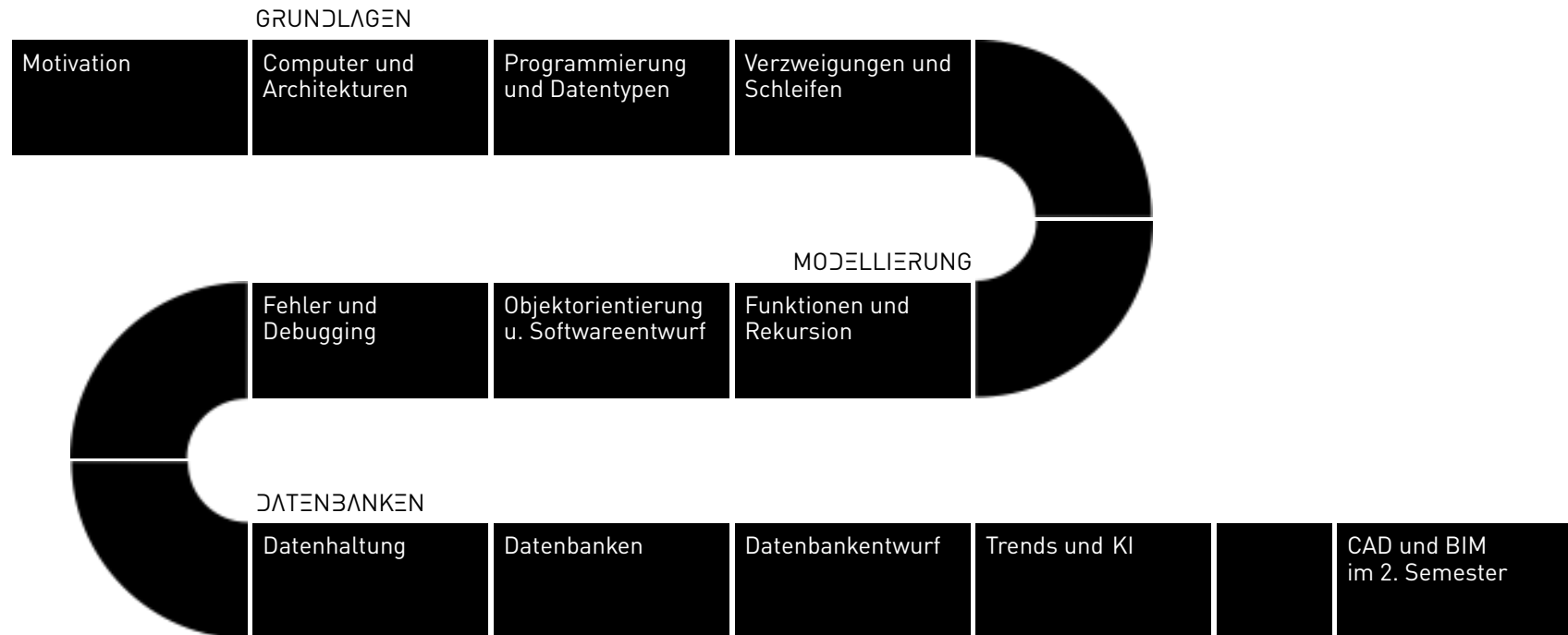


# TOPICS

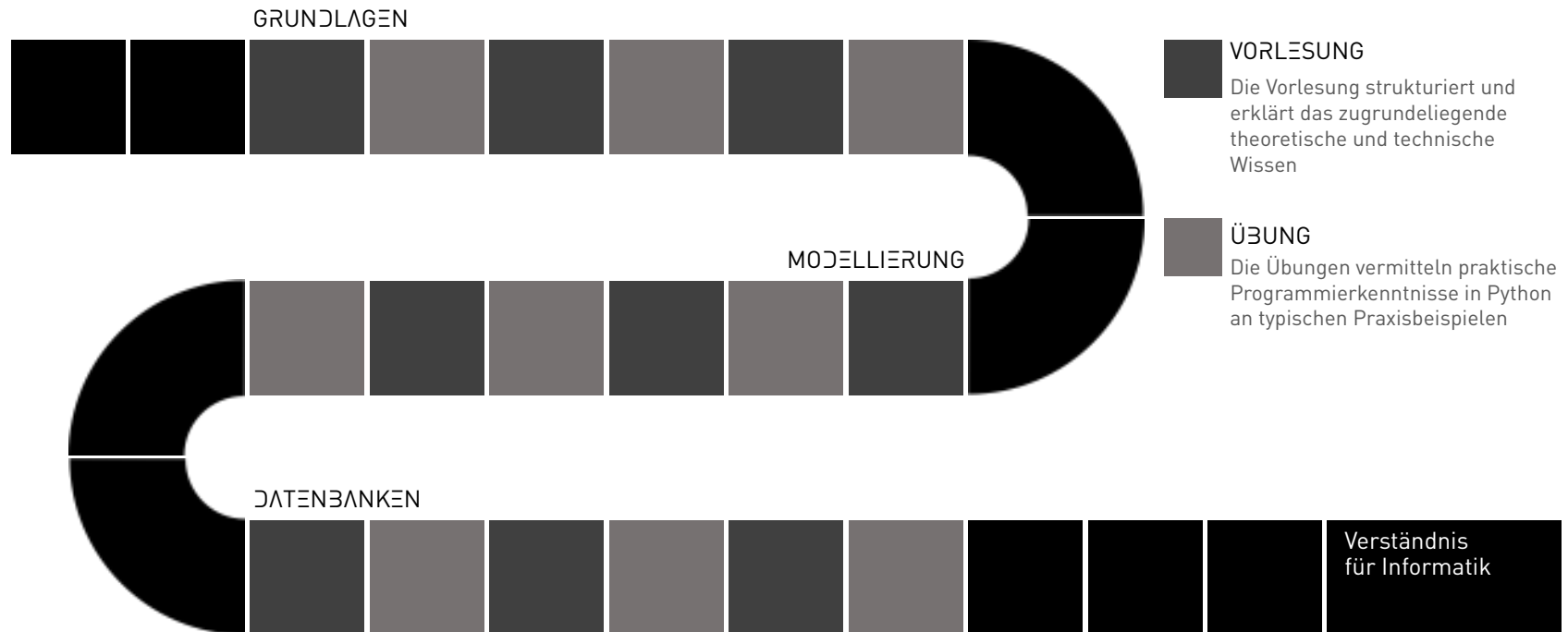




# PROCESS



# EXERCISES



# LECTURE SLIDES

Studienbegleitendes Portal für Präsenzlehre der Universität Rostock

Suche im WS 2023/24

STUD.IP

Overview Administration Participants Files Schedule References More ...

Vorlesung/Übung: Bau- und Umweltinformatik 1: Einführung in die Programmierung und Datenbanken - Files

Vorlesung/Übung: Bau- und Umweltinformatik 1: Einführung in die Programmierung und Datenbanken

| Type                     | Name                        | Size                 | Author         | Date              | Actions |
|--------------------------|-----------------------------|----------------------|----------------|-------------------|---------|
| <input type="checkbox"/> | 01b_Anwendungen.pdf         | 8.3 MB (0 Downloads) | Plönnigs, Jörn | 15/10/23 21:06:51 | ⋮       |
| <input type="checkbox"/> | 01c_Computerhardware.pdf    | 2.8 MB (0 Downloads) | Plönnigs, Jörn | 15/10/23 21:06:51 | ⋮       |
| <input type="checkbox"/> | 01d_Softwarearchitektur.pdf | 2 MB (0 Downloads)   | Plönnigs, Jörn | 15/10/23 21:06:51 | ⋮       |
| <input type="checkbox"/> | 01e_Programmiersprachen.pdf | 2.4 MB (0 Downloads) | Plönnigs, Jörn | 15/10/23 21:06:51 | ⋮       |

Download Move Copy Delete New folder Add file

Actions

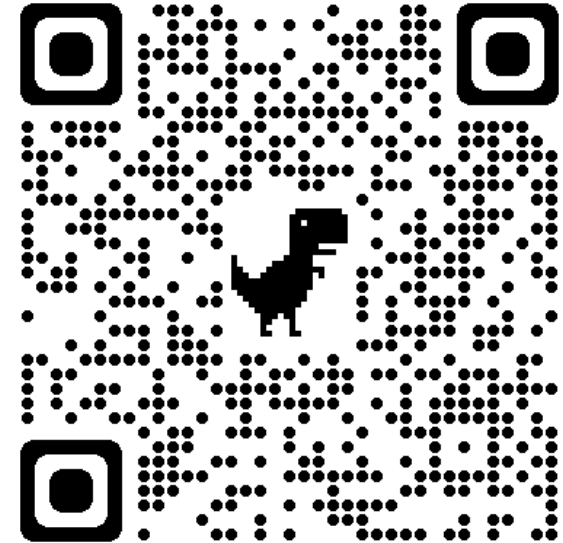
New folder Add file

Upload files

Add files via drag and drop to this area.

Views

Folder view All documents



# LECTURE DOCUMENTATION

AI4 SUSTAINABLE  
CONSTRUCTION

## Einführung

### Grundlagen

Grundlagen  
Computerhardware  
Softwarearchitekturen  
Programmiersprachen  
Wissenspyramide  
Quiz - Wissenspyramide  
Datentypen  
Operatoren  
Quizz: Boolesche  
Operationen  
Verzweigungen  
Schleifen

### Modularisierung

Funktionen  
Objekte  
Rekursive Funktionen  
Programmablauf  
Exceptions  
Unit-Tests  
Debugging  
Pakete

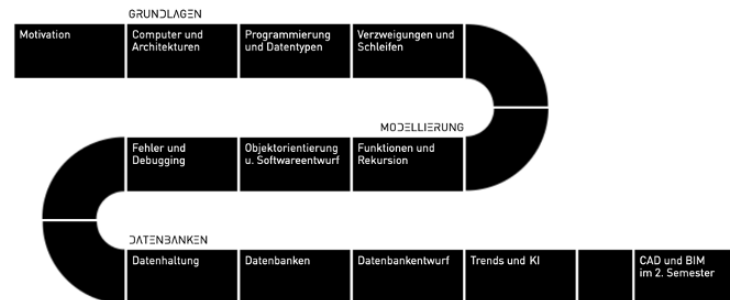
### Datenbanken

Umgang mit Dateien



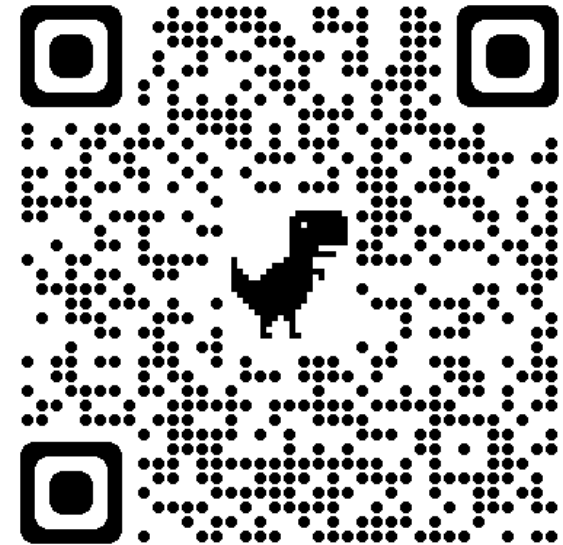
## Einführung

In diesem Kurs steht die Aus- und Weiterbildung von Informatikfähigkeiten speziell für den Anwendungsbereich der Bau- und Umweltingenieurwissenschaften im Fokus. Durch die Digitalisierung benötigen Ingenieure und andere technisch Interessierte in diesen Branchen immer häufiger grundlegende Fähigkeiten in der Programmierung, speziell im Bereich des Datenmanagements, -verarbeitung und -analyse. In den Bau- und Umweltingenieurwissenschaften handelt es sich hier z.B. um Datensätze aus Sensornetzwerken, Planungsprozessen oder Gebäudedatenbanken. Sehr gebräuchlich für die Arbeit mit solchen Daten ist die Programmiersprache Python mit ihren vielfältigen Programmbibliotheken. Das Bildungsangebot führt daher in die Grundlagen der Programmierung und den Umgang mit Datenbanken und Analysetools in Python ein. Es werden dabei zusätzlich theoretische Informatik-Grundlagen sowie fachspezifische Entwicklungsaspekte an verschiedenen Beispielen vermittelt (z.B. der Umgang mit räumlichen Daten).

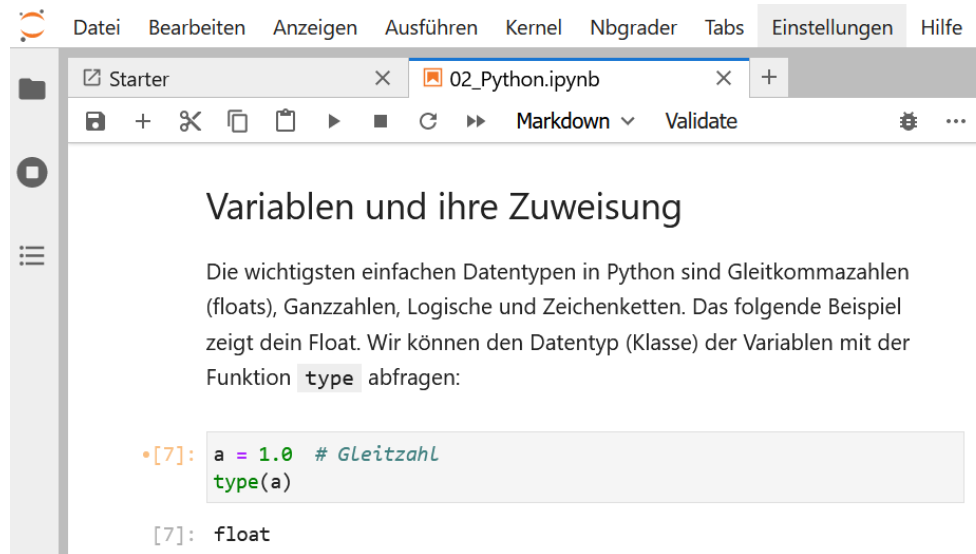


Der Kurs gliedert sich in drei Teile:

1. **Grundlagen:** Bietet eine Einführung in Hard- und Software von Computern und in die Programmierung in Python.
2. **Modellierung:** Thematisiert die Modellierung von Problemen mit Funktionen und Objekten und die Lösung von Problemen durch Debugging
3. **Datenbanken:** Gibt einen Überblick in das Management kleiner und größerer Daten durch Dateien und Datenbanken.



# EXERCISES



The screenshot shows a Jupyter Notebook window with the title '02\_Python.ipynb'. The menu bar includes 'Datei', 'Bearbeiten', 'Anzeigen', 'Ausführen', 'Kernel', 'Nbgrader', 'Tabs', 'Einstellungen', and 'Hilfe'. The toolbar contains icons for saving, adding, deleting, and running code, along with 'Markdown' and 'Validate' buttons. The notebook content is in Markdown format, titled 'Variablen und ihre Zuweisung'. The text explains that the simplest data types in Python are floats, integers, booleans, and strings, and provides an example of checking the type of a float variable using the `type` function. The code cell shows the assignment `a = 1.0` with a comment `# Gleitzahl` and the command `type(a)`. The output of the cell is `[7]: float`.

```
•[7]: a = 1.0 # Gleitzahl
      type(a)

[7]: float
```

Starting next week

- Office hours (PC Pool 1)  
Wednesday 11:15 AM – 12:45 PM
- Digital exercises:
  - Online environment
  - Tutorial videos

# EXERCISE



Midjourney: Repeat

## INTRODUCTION



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Phone: 0381 498-3503



# OBJECTIVES

Programming concepts learned in the lecture

- apply them repeatedly in practice
- apply them to other problems



Midjourney: Student aiming for a goal

## OBJECTIVE 2

Why learn programming as an engineer?

- Understanding the basic elements of programming
- Knowledge of the typical steps in a software project
- Experience in finding solutions
- Learn strategies for addressing problems
- In the later stages: understand the systematic nature behind software.
- Master the basics — from which you can continue to build!

## PROCESS - TWO PARTS

### *Tutorial video*

- Watch
- Code along to learn the approaches
  
- Both parts must be submitted!
- Group work is not planned!

### *Assignment*

- Think up your own solutions or research them

## CONSULTATION

Questions for: Markus Berger

Assistance with submissions: Clemens Kujat (TA)

Starting next week

[Markus.Berger@uni-rostock.de](mailto:Markus.Berger@uni-rostock.de)

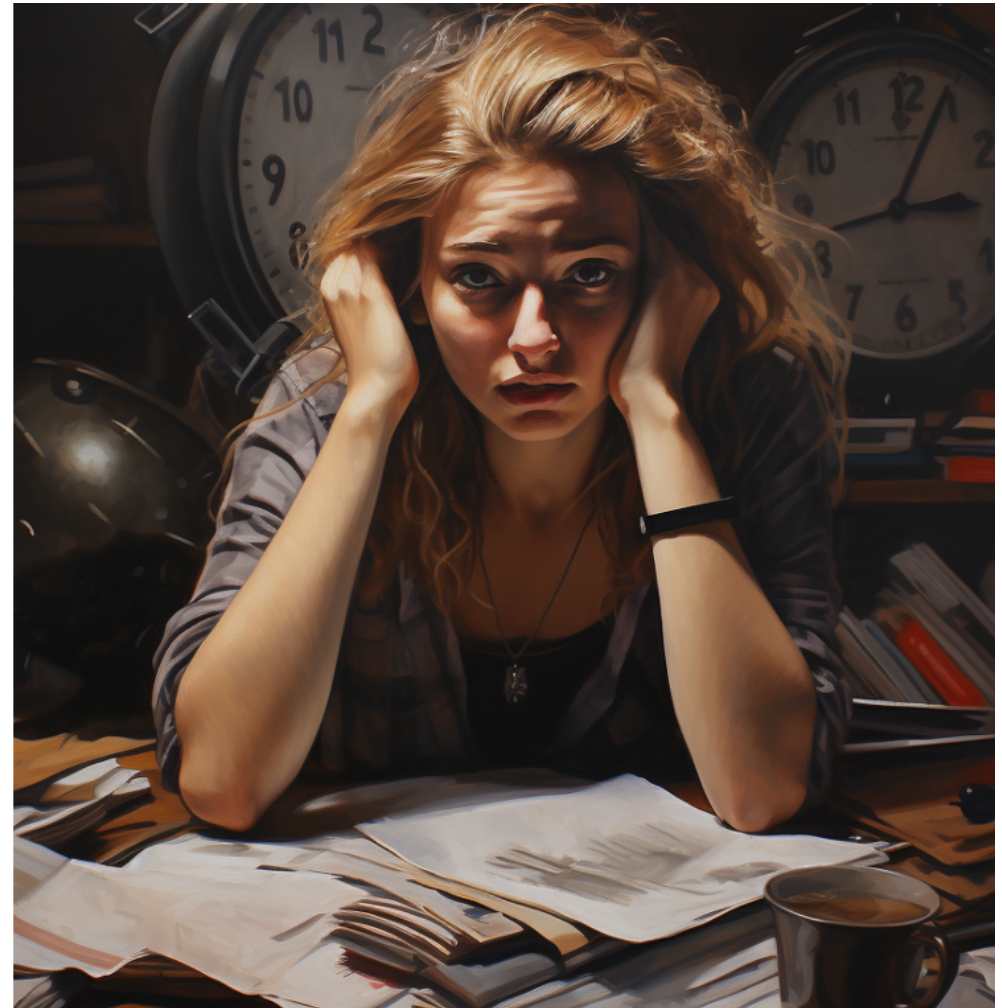
[Clemens.kujat@uni-rostock.de](mailto:Clemens.kujat@uni-rostock.de)

- Wednesday 11:15–12:45 in PC-Pool 1 and via Zoom
- Clarification of questions about the videos and the assignments

# SUBMISSION

Practice videos:

- Video releases every Tuesday
- Consultations every Wednesday
- Submissions due every Monday



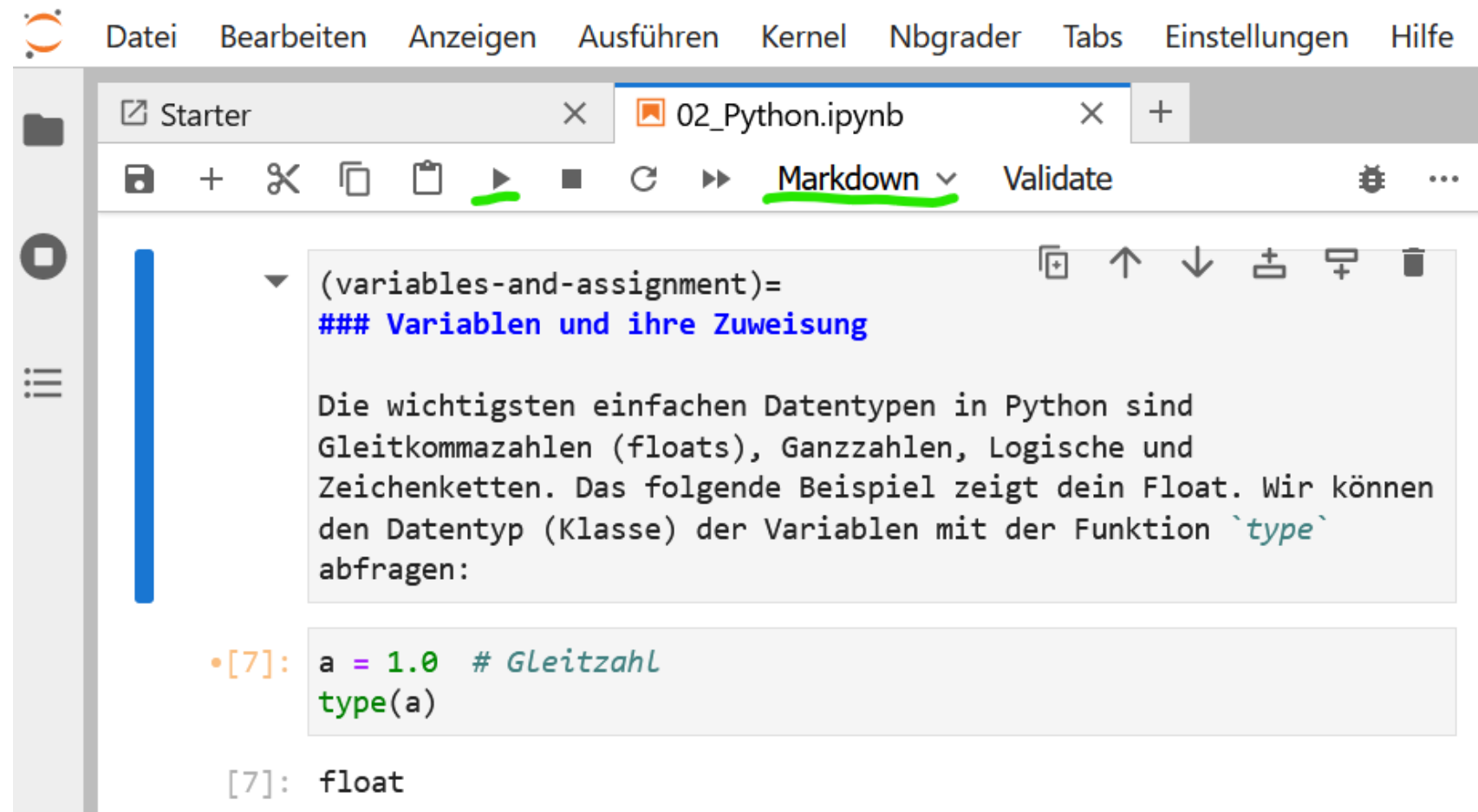
Midjourney: Student with a deadline

## EXAM ELIGIBILITY

- Homework assignments are not graded
- Instead, they are graded as Met / Not Met for the exercise as a whole (no partial submissions)
- By the end of the semester, at least 50% of the exercises must be completed
- More extensive tasks carry a larger share of the grade

| Topic                           | Weighting |
|---------------------------------|-----------|
| <b>Foundations</b>              |           |
| Python & Data Types             | 5%        |
| Operators, Conditionals & Loops | 5%        |
| Functions & Objects             | 5%        |
| Algorithms                      | 5%        |
| <b>Advanced</b>                 |           |
| Errors & Tests                  | 15%       |
| Design                          | 15%       |
| Data Management                 | 20%       |
| <b>Advanced</b>                 |           |
| Database Queries                | 15%       |
| Database Design                 | 15%       |

# ONLINE PYTHON IDE - JUPYTER BOOKS



The screenshot displays the JupyterLab web interface. At the top, a menu bar includes 'Datei', 'Bearbeiten', 'Anzeigen', 'Ausführen', 'Kernel', 'Nbgrader', 'Tabs', 'Einstellungen', and 'Hilfe'. Below the menu, a tab bar shows two open notebooks: 'Starter' and '02\_Python.ipynb'. The '02\_Python.ipynb' notebook is active, showing a toolbar with icons for saving, adding, deleting, and running cells, along with a 'Markdown' dropdown menu and a 'Validate' button. The notebook content is divided into two sections. The first section is a markdown cell with the title '(variables-and-assignment)= ### Variablen und ihre Zuweisung' and a paragraph explaining basic Python data types (floats, integers, booleans, and strings) and how to check a variable's type using the 'type' function. The second section is a code cell containing the code 'a = 1.0 # Gleitzahl' and 'type(a)', which has been executed, resulting in the output '[7]: float'.

File Edit View Execute Kernel Nbgrader Tabs Settings Help

Starter 02\_Python.ipynb

Save Add Delete Run Markdown Validate

(variables-and-assignment)=  
### Variablen und ihre Zuweisung

Die wichtigsten einfachen Datentypen in Python sind Gleitkommazahlen (floats), Ganzzahlen, Logische und Zeichenketten. Das folgende Beispiel zeigt dein Float. Wir können den Datentyp (Klasse) der Variablen mit der Funktion `type` abfragen:

```
•[7]: a = 1.0 # Gleitzahl  
      type(a)
```

```
[7]: float
```

## SIGN UP FOR JUPYTER & SUBMISSION

- Create the account in the coming week at:  
<https://ml-lab.ai4sc-lectures.auf.uni-rostock.de/>
- With the following username: 'firstname\_lastname'
- An introduction will then be in the first exercise video

**Sign In**

**Username:**

**Password:**  
 

**Sign In**

[Sign up](#) to create a new user.



## NOTE: AI

- Powerful tools for programming – when you know how to use them
- A basic understanding of Python goes a long way in using these tools
- At the end, an extra lecture and exercise on the topic
- But: During the exam you must demonstrate a solid understanding of programming, so the exercises must be completed by you yourselves!

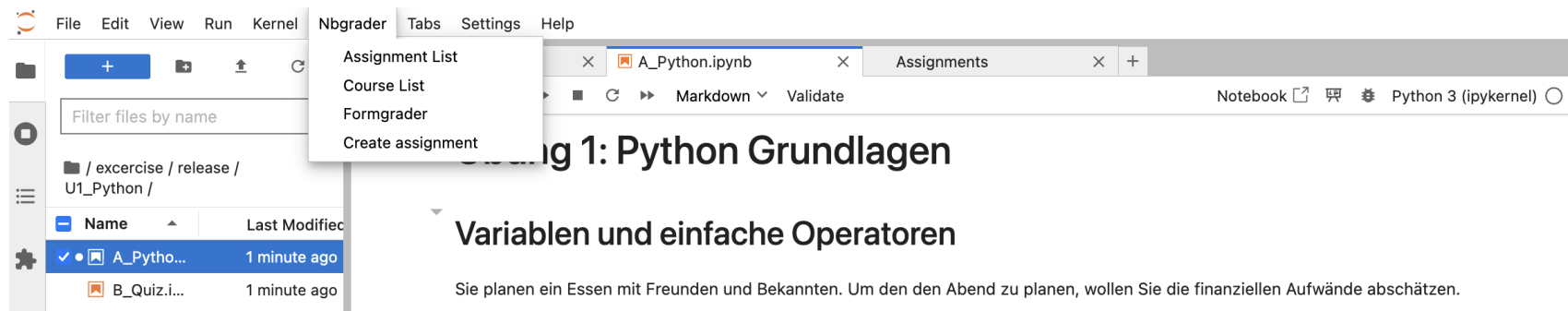
# HOMEWORK

©Midjourney: Programming in Python

# FETCHING TASKS

Assignments can be downloaded, edited, and submitted on the JupyterLab platform

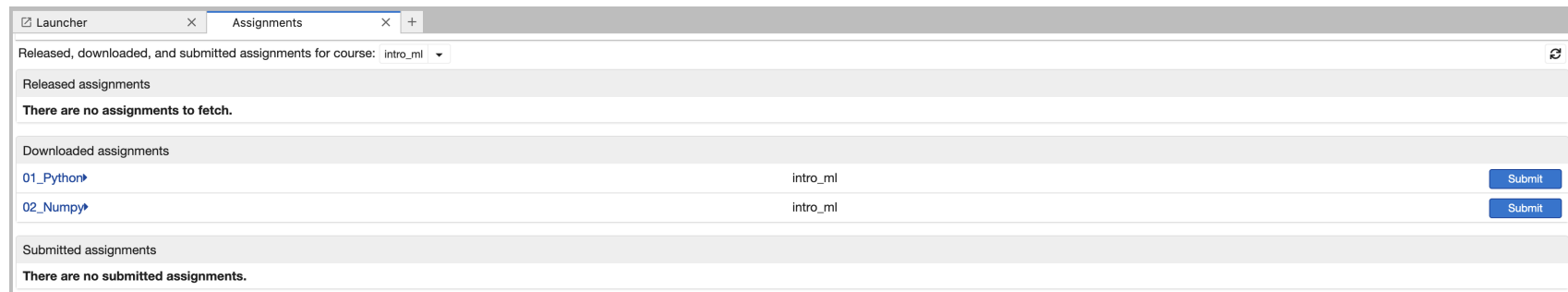
- Menu item **Nbgrader/Assignments** → the main option, showing the page with the assignments
- Menu item **Nbgrader/Courses** → lists all enrolled courses
- Menu item **Nbgrader/Formgrader** → only for instructors, for viewing and grading student submissions



- Menu item `Nbgrader/Assignments` → the current assignments view opens
- Using `fetch` → download the selected assignment



- After downloading, the assignment appears in the "Downloaded Assignments" list
- Click the blue assignment name → opens the assignment's subfolder
- In the subfolder: all notebooks and related files are visible



## EDITING TASKS

- Read the task description.
- Replace `YOUR CODE HERE` with the correct code.
- Replace `YOUR ANSWER HERE` with a text answer.
- Delete the `raise NotImplementedError()` if applicable.
- Run the cell with `Shift+Enter`.
- Ensure the execution was successful.

```
[ ]: # test bezahlbar
assert callable(bezahlbarkeit)
bezahlbar = bezahlbarkeit(budget, gesamtpreis)
assert bezahlbar == "Ich kann es mir leisten" if budget >= gesamtpreis else bezahlbar == "Ich kann es mir nicht leisten"
print("Richtig!")
```

- For every code cell that needs to be completed, a test cell follows.
- These cannot be edited.
- But running them will show you whether they are correct.

```
[ ]: # test bezahlbar
assert callable(bezahlbarkeit)
bezahlbar = bezahlbarkeit(budget, gesamtpreis)
assert bezahlbar == "Ich kann es mir leisten" if budget >= gesamtpreis else bezahlbar == "Ich kann es mir nicht leisten"
print("Richtig!")
```

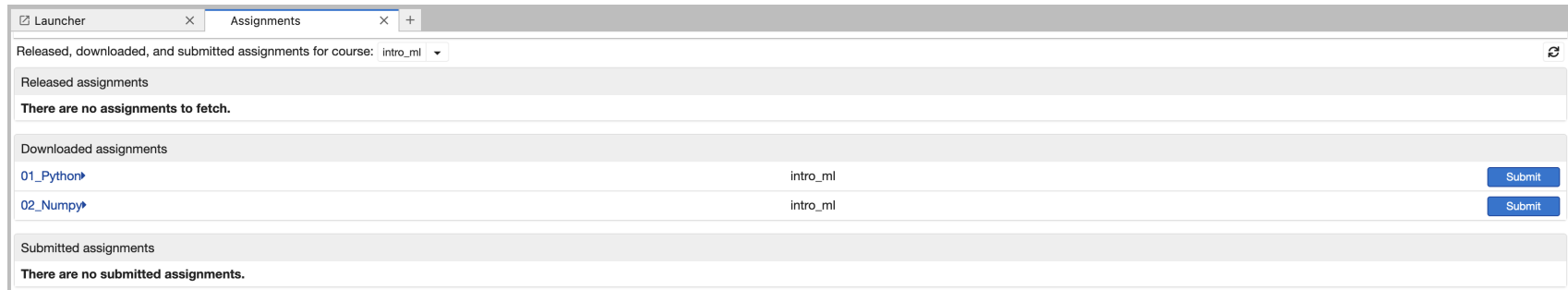
## SUBMITTING ASSIGNMENTS

- Validate every notebook before submission
- Use the **Validation** button in the menu to start the validation
  - Successful validation → message: “Success! Your notebook passes all the tests.”
  - Failed validation → a report of the errors that occurred is displayed





- After successful validation of all notebooks → press the 'Submit' button
- The assignment is submitted
- The submitted assignment appears in the "Submitted Assignments" list



## SHUT DOWN THE SERVER

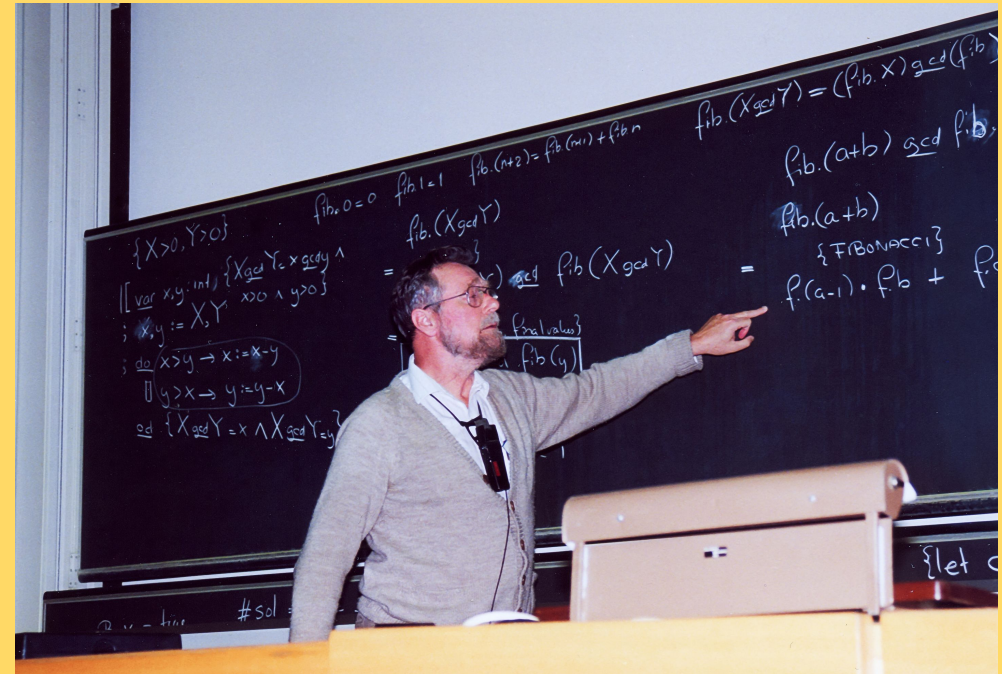
- Shut down the JupyterLab environment properly when you are finished with your work
- Use the menu item `File/Hub Control Panel` to switch to the Server Control view
- Shut down the server with `Stop my Server`

# MOTIVATION

©Midjourney: Programming in Python

## LECTURE HALL QUESTION

What is computer science?



"Computer science is no more about computers than astronomy is about telescopes." —  
Edsger W. Dijkstra

## DEFINITION - COMPUTER SCIENCE

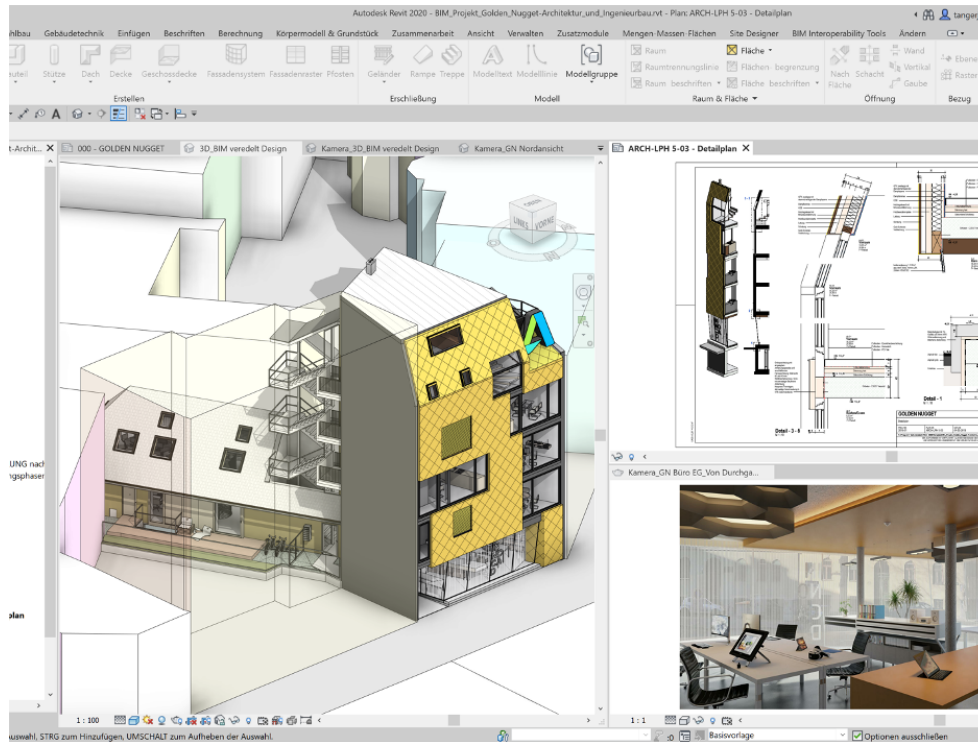
### **Definition: Computer Science**

Computer science is the science of the systematic processing of information.  
Origin: Mathematics (logic, algorithms) + Electrical Engineering (hardware)

## WHERE DO ENGINEERS NEED COMPUTER SCIENCE?

- Computer-Aided Design
- Virtual environments
- Structural engineering
- ...
- Construction process automation
- Acceptance measurements
- Construction robotics
- ...
- Geographic Information Systems
- Facility management
- Building automation
- ...

# DRAFT: CAD - COMPUTER AIDED DESIGN



Software applications for designing, drafting, and presenting engineering drawings and plans, for both 2D and 3D models. (SoftSelect Glossary / Lexicon)

## DRAFT: VIRTUAL ENVIRONMENTS

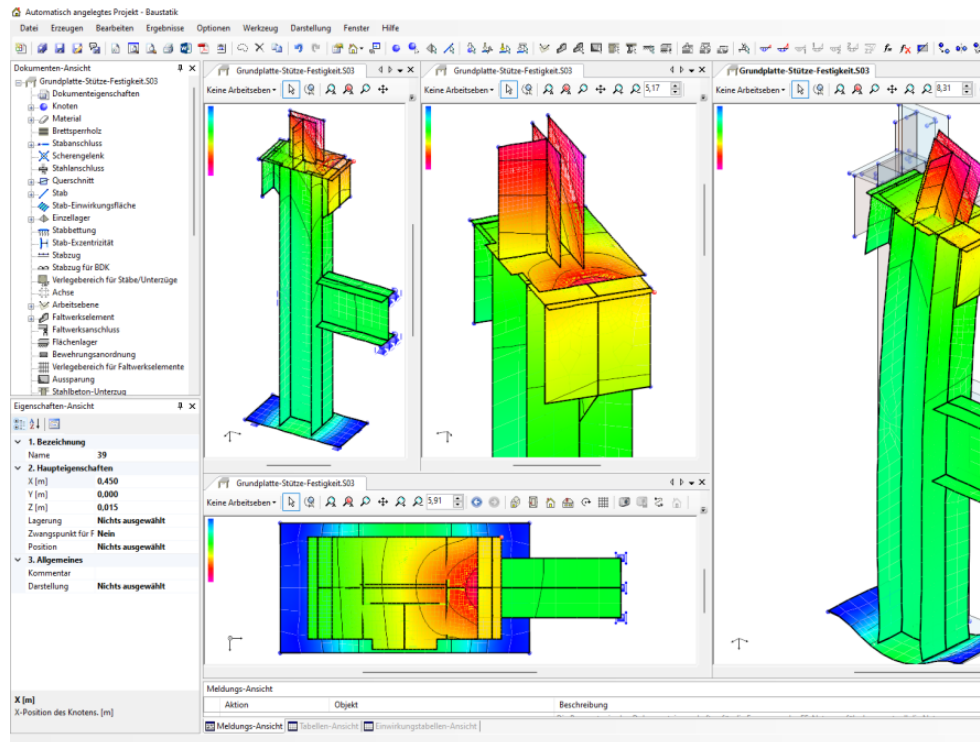
Applications that allow visualizing and exploring designs in VR (Virtual Reality) or AR (Augmented Reality).





# DESIGN: STRUCTURAL ENGINEERING

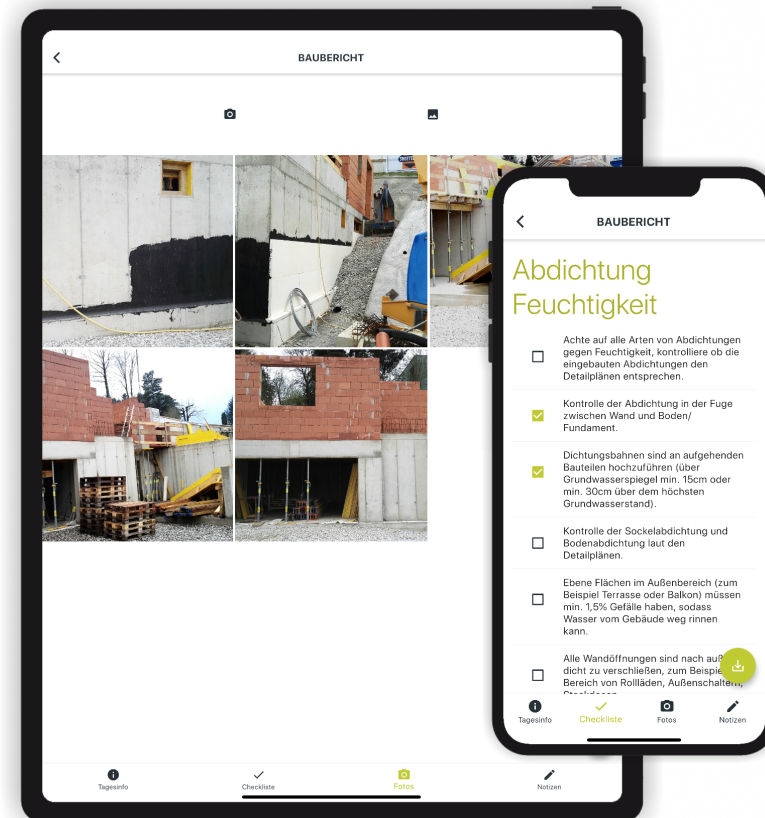
The aim of his structural design is to balance the required load-bearing capacity and serviceability of a building structure during its intended service life with the demands of economy and aesthetics. [German Wiki](#)



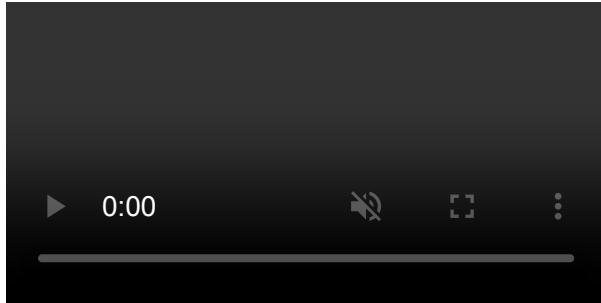
# CONSTRUCTION: AUTOMATING THE CONSTRUCTION PROCESS

Automation of various steps

- Daily construction log
- Construction scheduling
- Construction project management



## CONSTRUCTION: CONSTRUCTION ROBOTICS

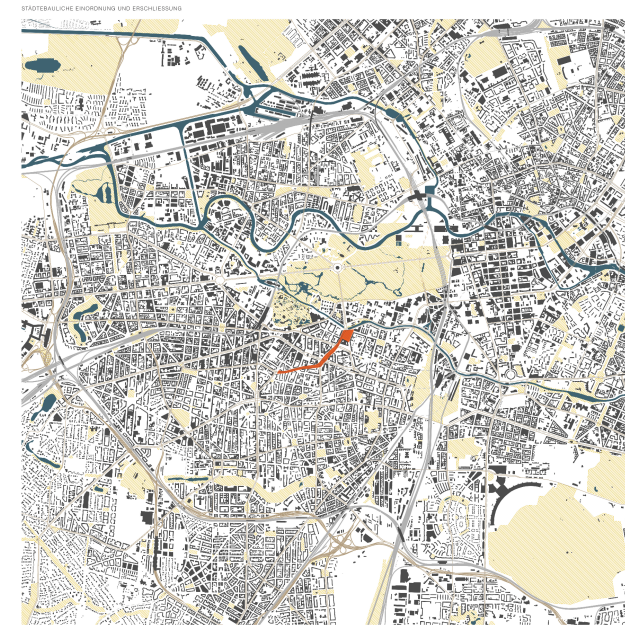
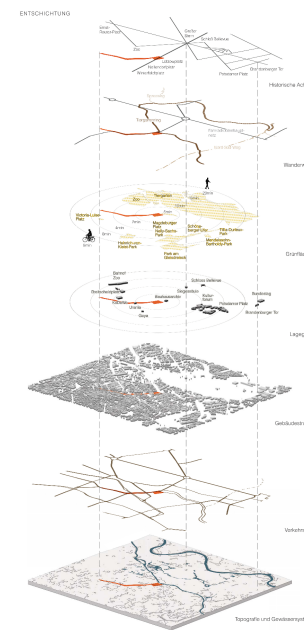


The scope of application for robots in the construction industry is fundamentally independent of the construction-specific business model, i.e., independent of the segment, the activity, and the actual tasks involved.

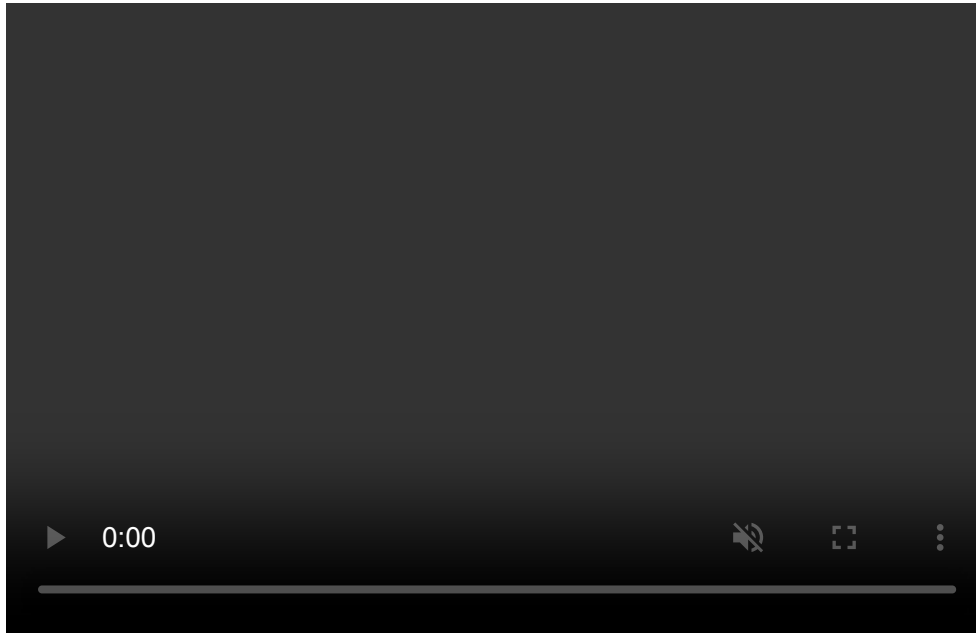
# OPERATIONS: GIS - GEOINFORMATION SYSTEMS

Geoinformation systems are information systems for the capture, processing, organization, analysis, and presentation of spatial data. [DeWiki](#)

## ANALYSE - STADTRAUM



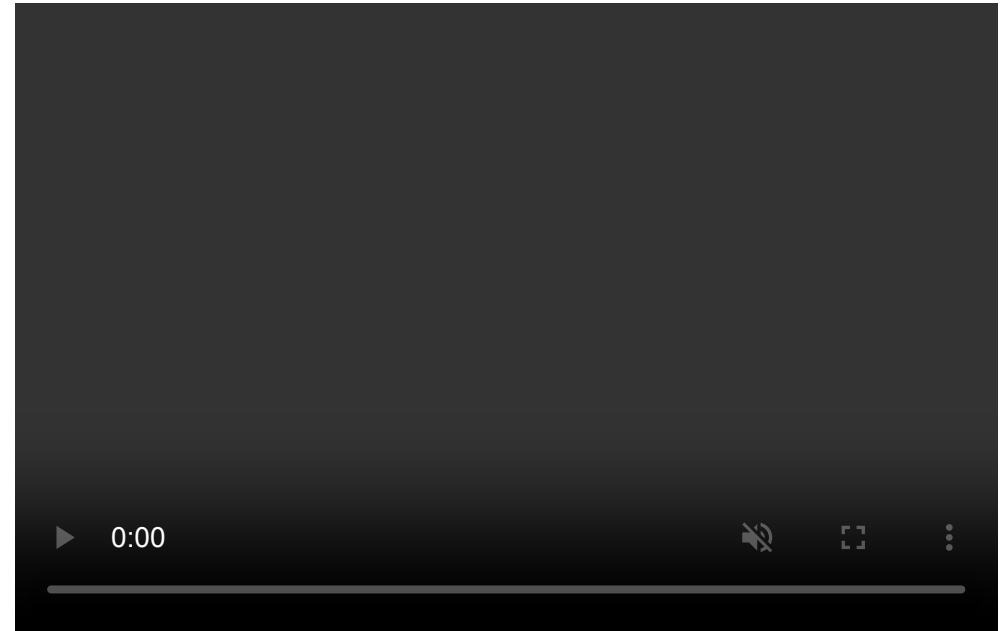
## OPERATION: BUILDING AUTOMATION



The totality of monitoring, control, regulation, and optimization systems in buildings. [DeWiki](#)

## OPERATION: CAFM - COMPUTER-AIDED FACILITY MANAGEMENT

Facility management refers to the administration and operation of buildings as well as their technical systems and facilities (English: facilities). CAFM is the computer-aided support of facility management. [DeWiki](#)



## LITERATURE RECOMMENDATIONS

- Python 3: The Comprehensive Handbook; Ernesti, Johannes, Kaiser, Peter, 2020
- Databases: Concepts and Languages; Saake, Gunter, Sattler, Kai-Uwe, Heuer, Andreas, 2018
- Recommendation: Use various online tutorials! <https://docs.python.org/3/tutorial/> & <https://www.w3schools.com/python/>
- In the long run, the most important thing: dive in at the deep end!

Questions?



